

From Tournament to Trading

... a brief tour of how we turn predictions into a Hedge Fund

agenda

- a day in the life
 - tournament
 - portfolio construction
 - trading
 - the aftermath
- Q&A

Admin! (1 of 2)

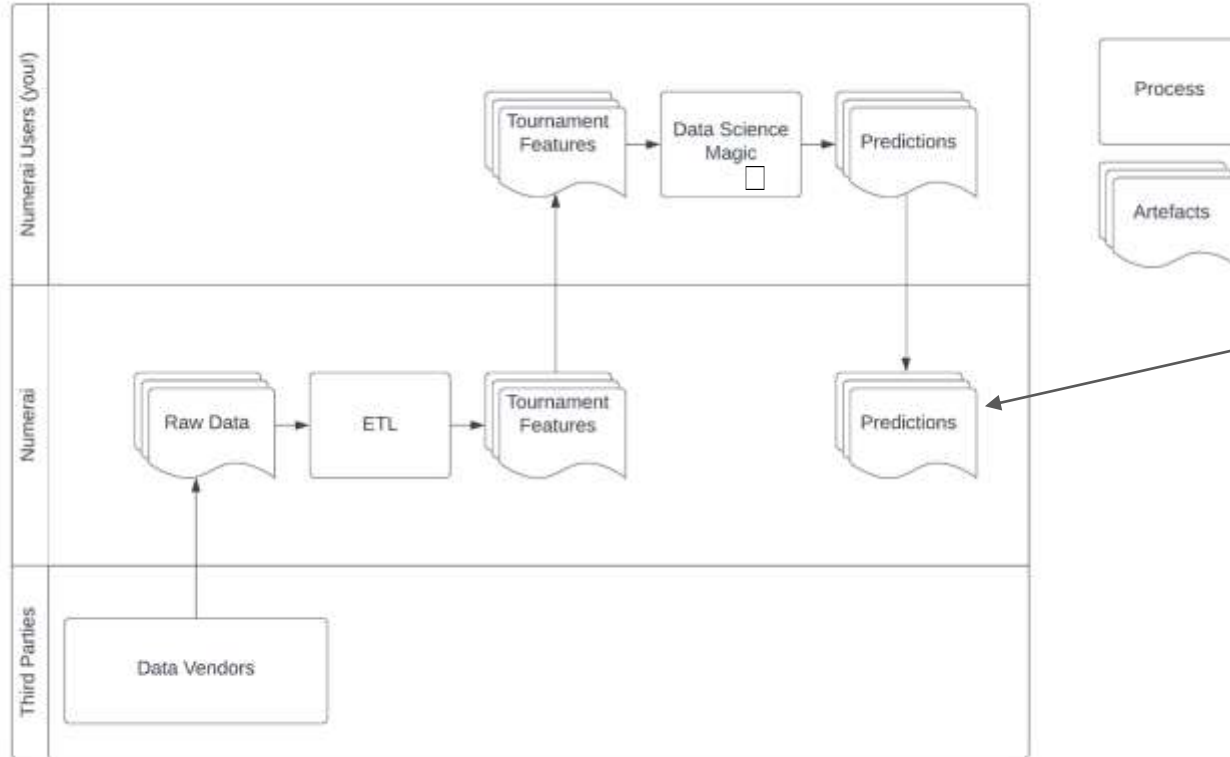
- Everything here is for educational purposes only!
 - i.e. not a representation to (prospective) investors
 - Subject to change without notice or announcement
- Stuff I'm not going to talk about:
 - TC (even though it's the future)
 - Performance
 - Supreme (except to say that we've started trading)

Admin! (2 of 2) who am I?

- Trading Engineering at Numerai
- Came from an HF options shop
- Most of my career spend in trading engineering at a CTA
- @jelford on github

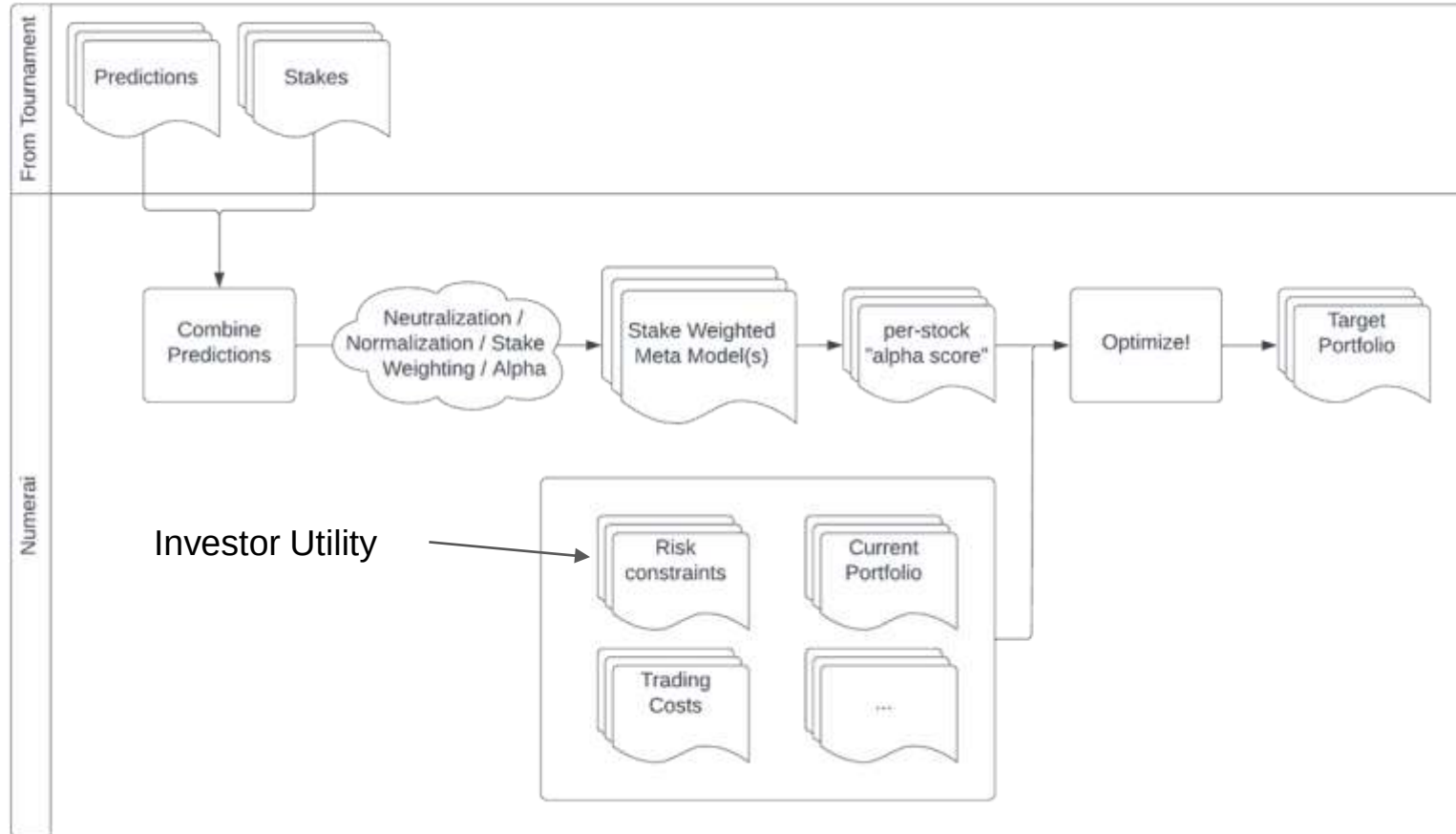
Okay now let's talk about trading

alpha generation

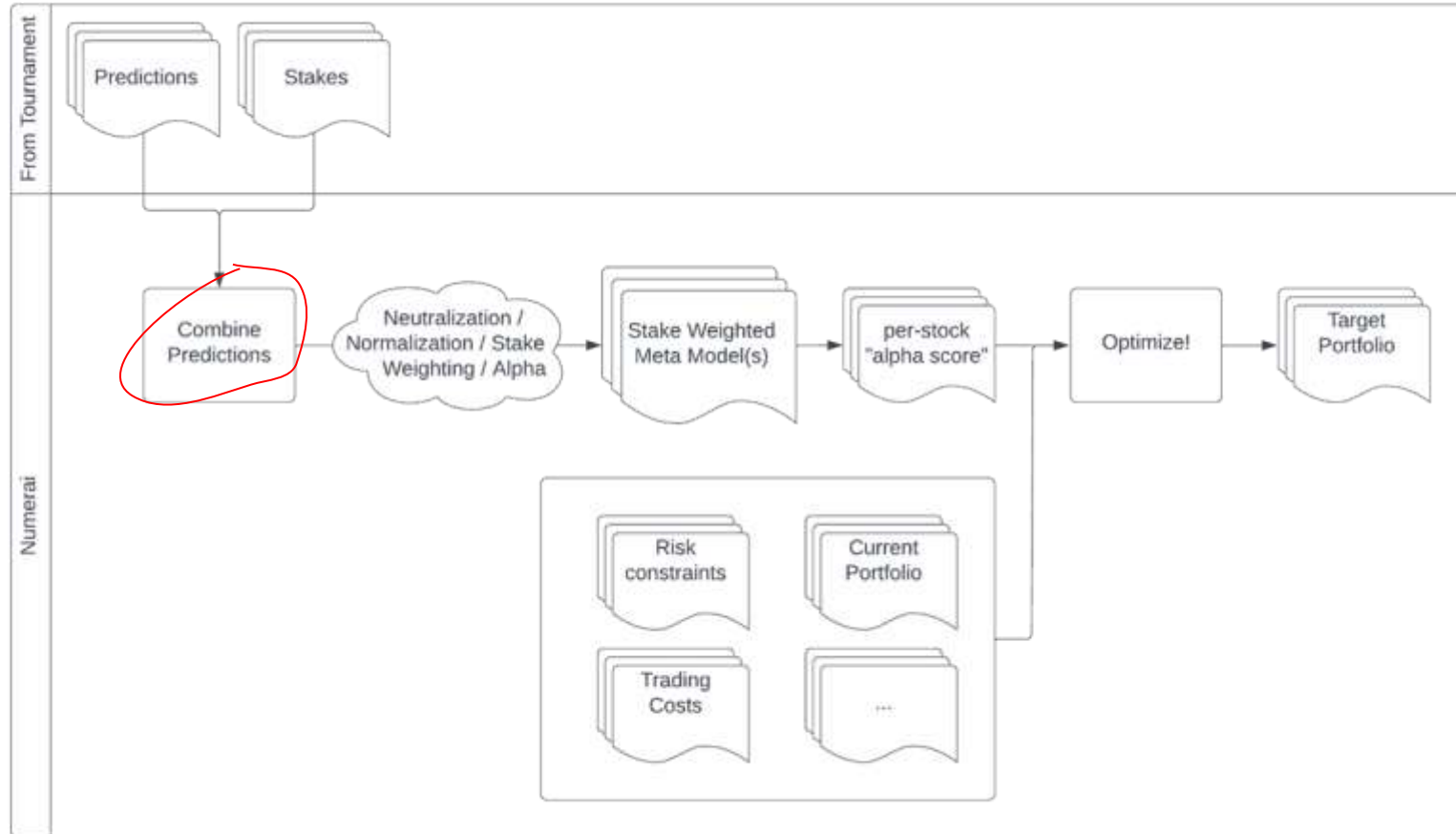


This is what we're after

incorporating knowledge into the portfolio



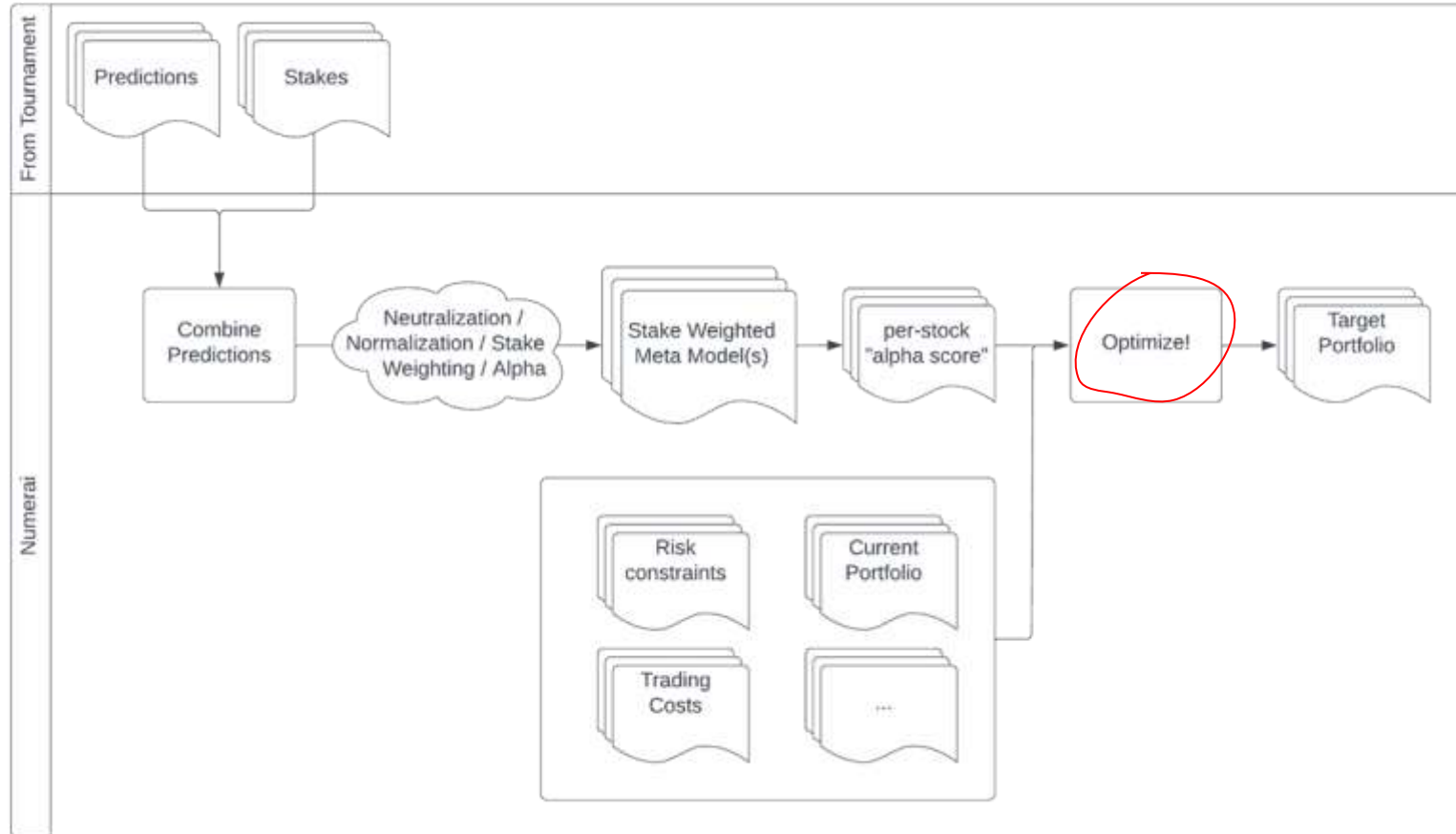
incorporating knowledge into the portfolio



stake weighting

```
predictions = user_preds @ user_stakes
```

incorporating knowledge into the portfolio



optimization

```
longs = cvx.Variable(num_stocks, pos=True)
shorts = cvx.Variable(num_stocks, pos=True)
```

```
interest_rate_exposure_penalty = cvx.sum_squares(
    longs @ interest_rate_exposures
    + shorts @ interest_rate_exposures
)
```

```
max_move_constraint = (longs - shorts) - (old_long - old_short) < max_move
```

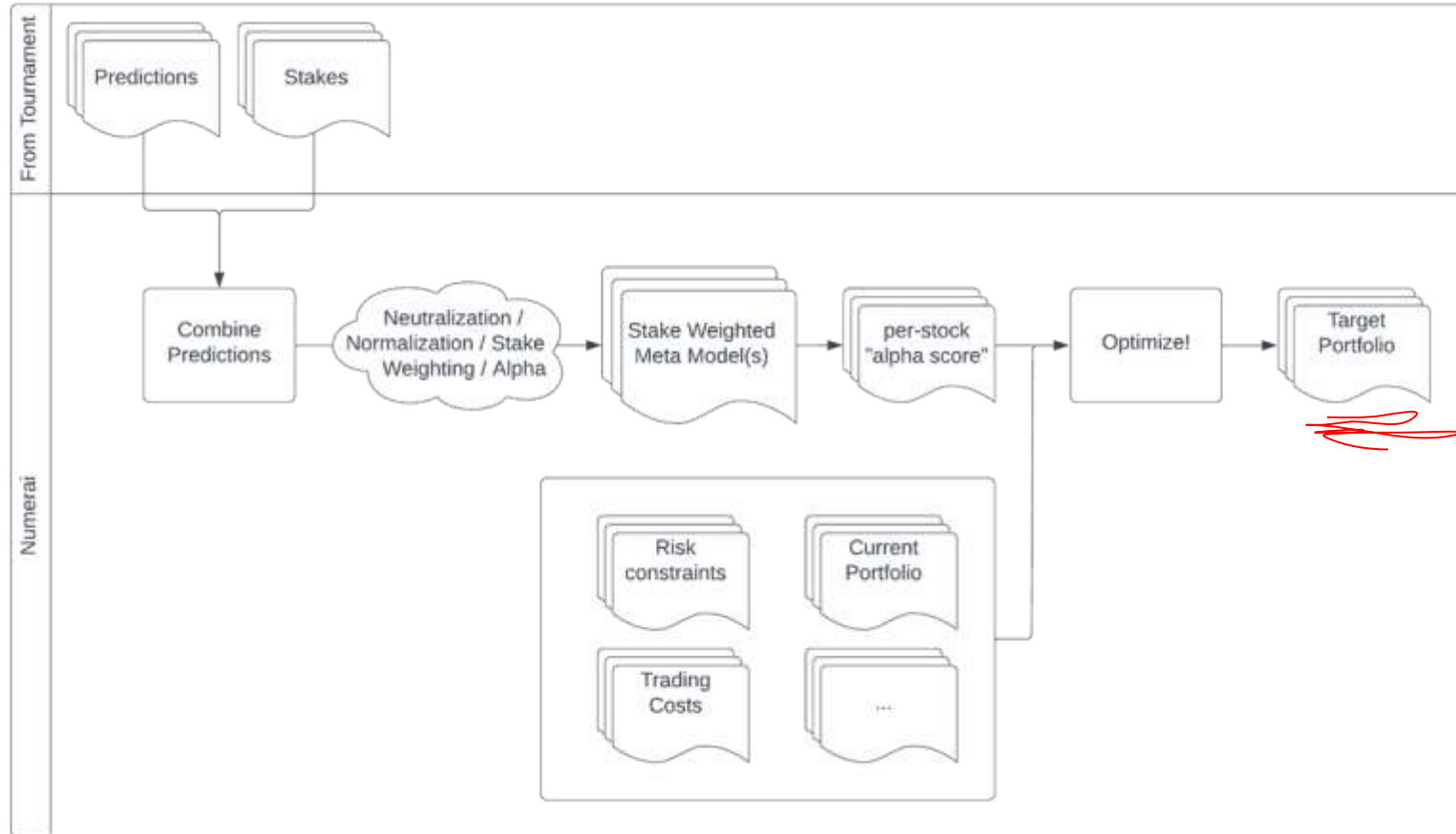
```
penalties = interest_rate_exposure_penalty # + a bunch of other penalties
constraints = max_move_constraint # + a bunch of other constraints
objective = scores @ (longs - shorts)
```

```
problem = cvx.Problem(
    cvx.Maximize(objective - penalties),
    constraints
)
```

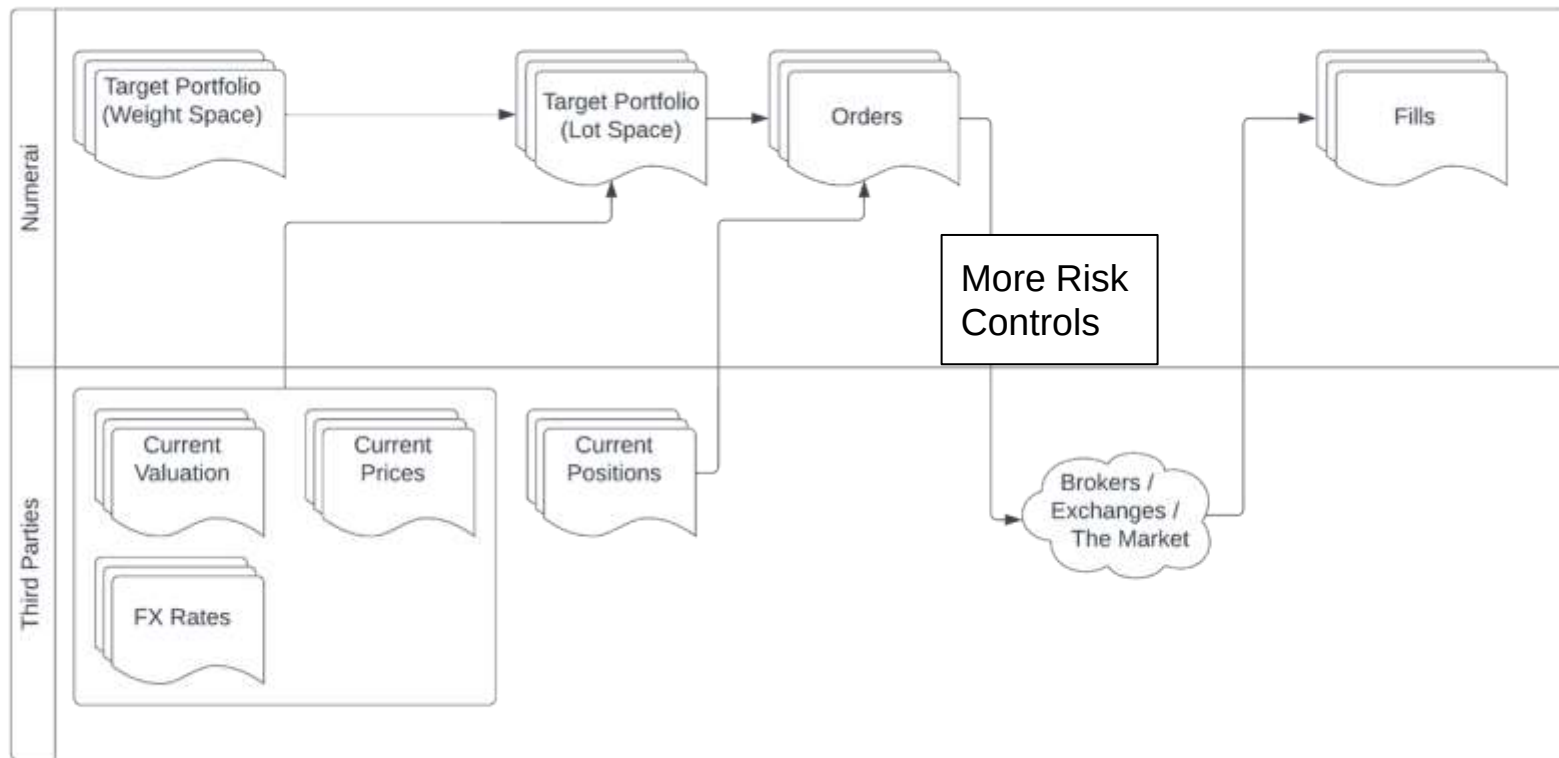
```
problem.solve()
```

```
return longs.value, shorts.value
```

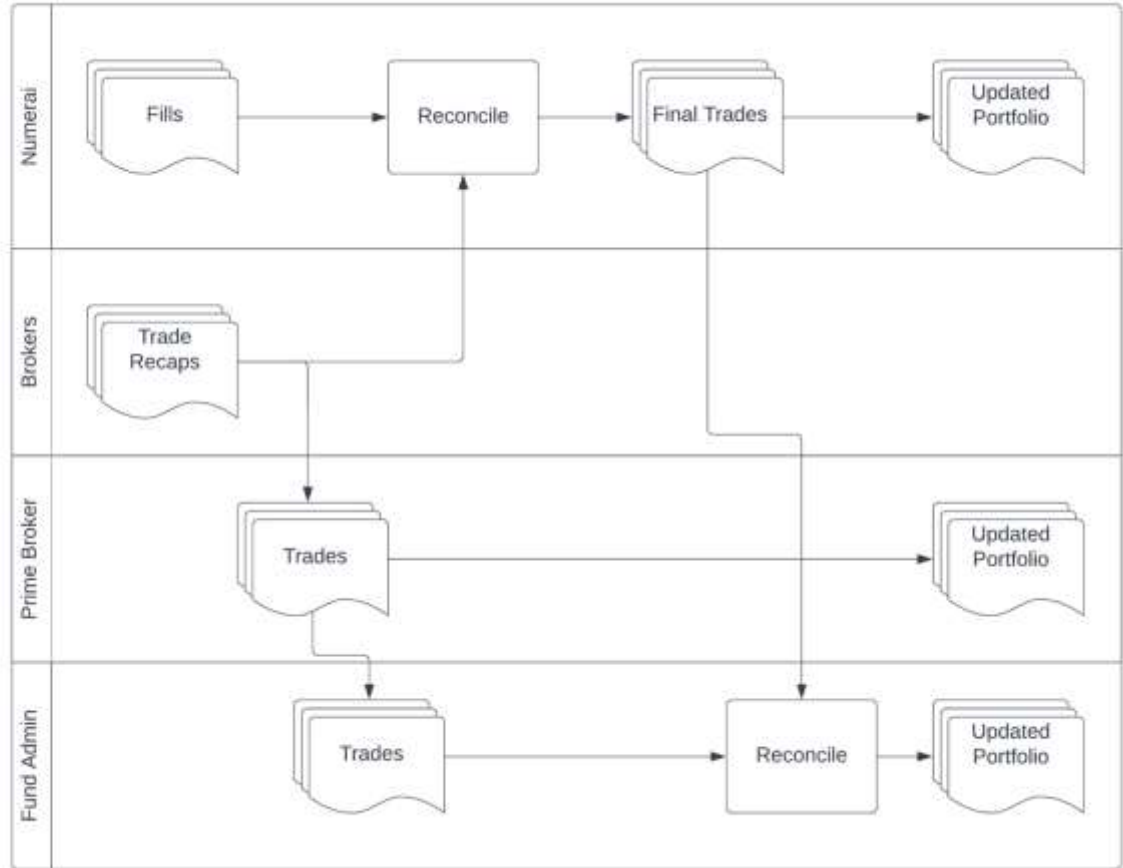
incorporating knowledge into the portfolio



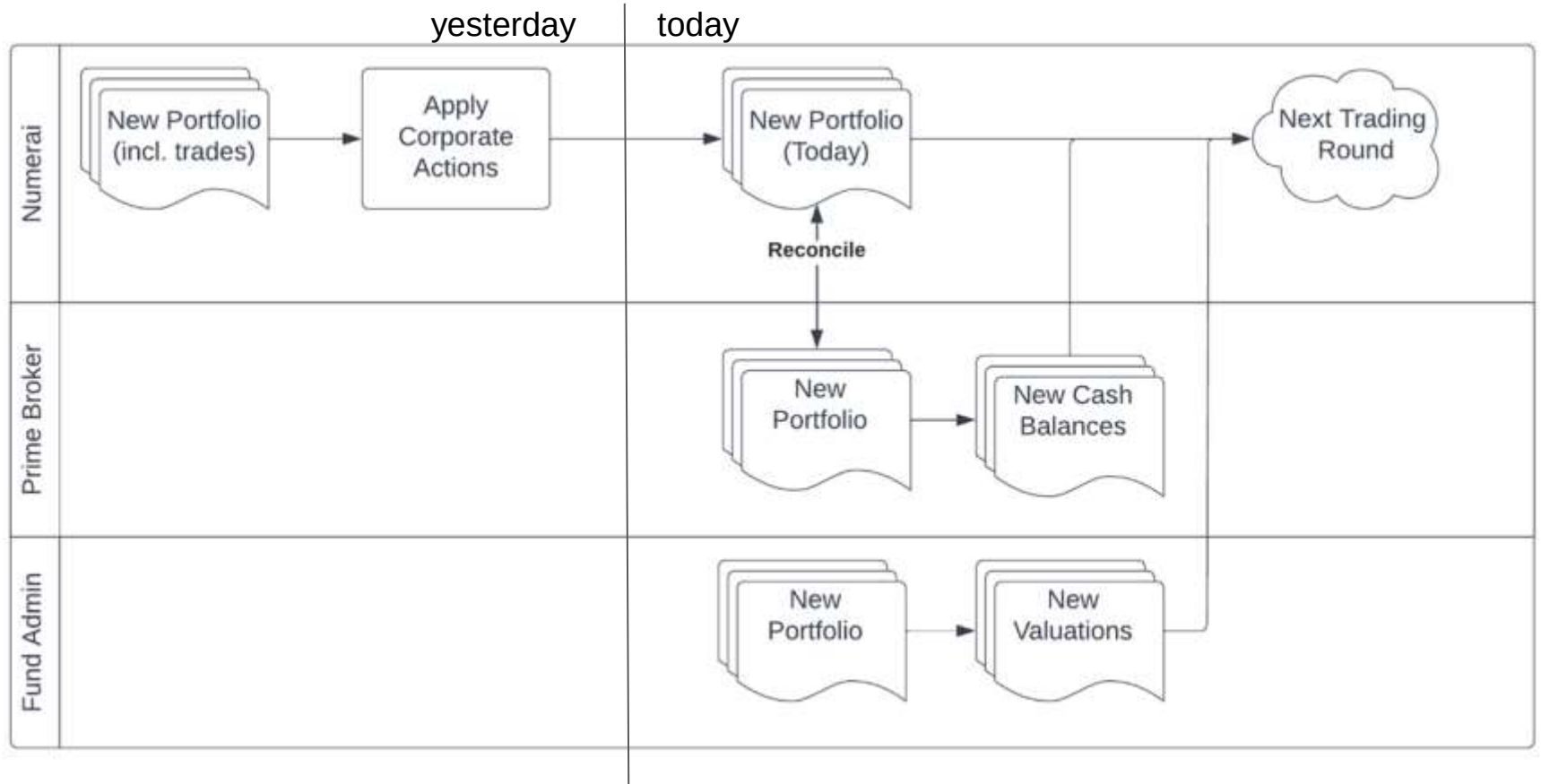
trading



the aftermath



the aftermath (cont)



what's new in trading?

Nothing very visible outside of Numerai, but we've been kind of busy?

- multi-fund support (supreme!)
- moved from R to Python
- risk controls in the trading chain
- reworking short locates process
- changes to exactly when we feature neutralize
- updates to how we expect prediction scores map to returns
- ...

AMA!

but just know that I might not be able to answer 🙋

+ btw we're hiring